BUS7C2 Finance and Accounting for Business

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Lecture 8: Analysing Financial Statement – Financial Ratios

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Learning Objective

At the end of this lecture; students will:

1. **Evaluate** the benefits and limitations of financial ratio analysis in strategic decision-making.
2. **Interpret** profitability, liquidity, efficiency, and gearing ratios using financial statements.
3. **Integrate** ratio insights with business context for effective performance assessment.

Financial Ratio Analysis

🔍 What is Ratio Analysis?

- *A **tool** that uses **quantitative relationships** between financial figures to assess a company's performance, financial health, and investment potential.*
- *Ratios **simplify the interactions** between financial variables, such as the relationship between sales, costs, and inventory.*

📁 Types of Financial Ratios

1. **Liquidity Ratios** – ability to meet short-term obligations
2. **Profitability Ratios** – earnings performance and margins
3. **Efficiency Ratios** – asset and resource usage
4. **Solvency Ratios** – long-term financial stability

💡 Why Use It?

- ***Compares businesses*** regardless of size or strategy
- ***Measures*** efficiency, profitability, liquidity, and solvency
- ***Supports better*** management and investment decisions

Everyday Ratios : Examples

💷 1. Monthly Budget vs. Spending

Imagine this: You plan to spend £300 on groceries but end up spending £250.

→ *This tells you how close you are to your plan.*

🟡 **It's like a financial control ratio** — *did you stick to your spending goals?*

🕒 2. Time Studied vs. Time Scrolled on Your Phone

You meant to study 3 hours but spent 1 hour on social media.

→ *Only two-thirds of your time was used effectively.*

🟡 **Similar to efficiency ratios** — how well are you using your resources?

🚙 3. Fuel in Your Tank vs. Distance You Drove

You filled your car with £30 of fuel and drove 250 miles.

→ *Are you getting good value for what you spent?*

🟡 **Like asset turnover** — how much result do you get from what you own?

💰 4. Earning vs. Saving Each Month

You earn £2,000 and save £200.

→ That's 10% saved.

🟡 **It's like a profit ratio** — how much are you keeping from what comes in?

Using Financial Ratios

Why Use Them?

They highlight trends and relationships in a business—but on their own, they need context to be useful.

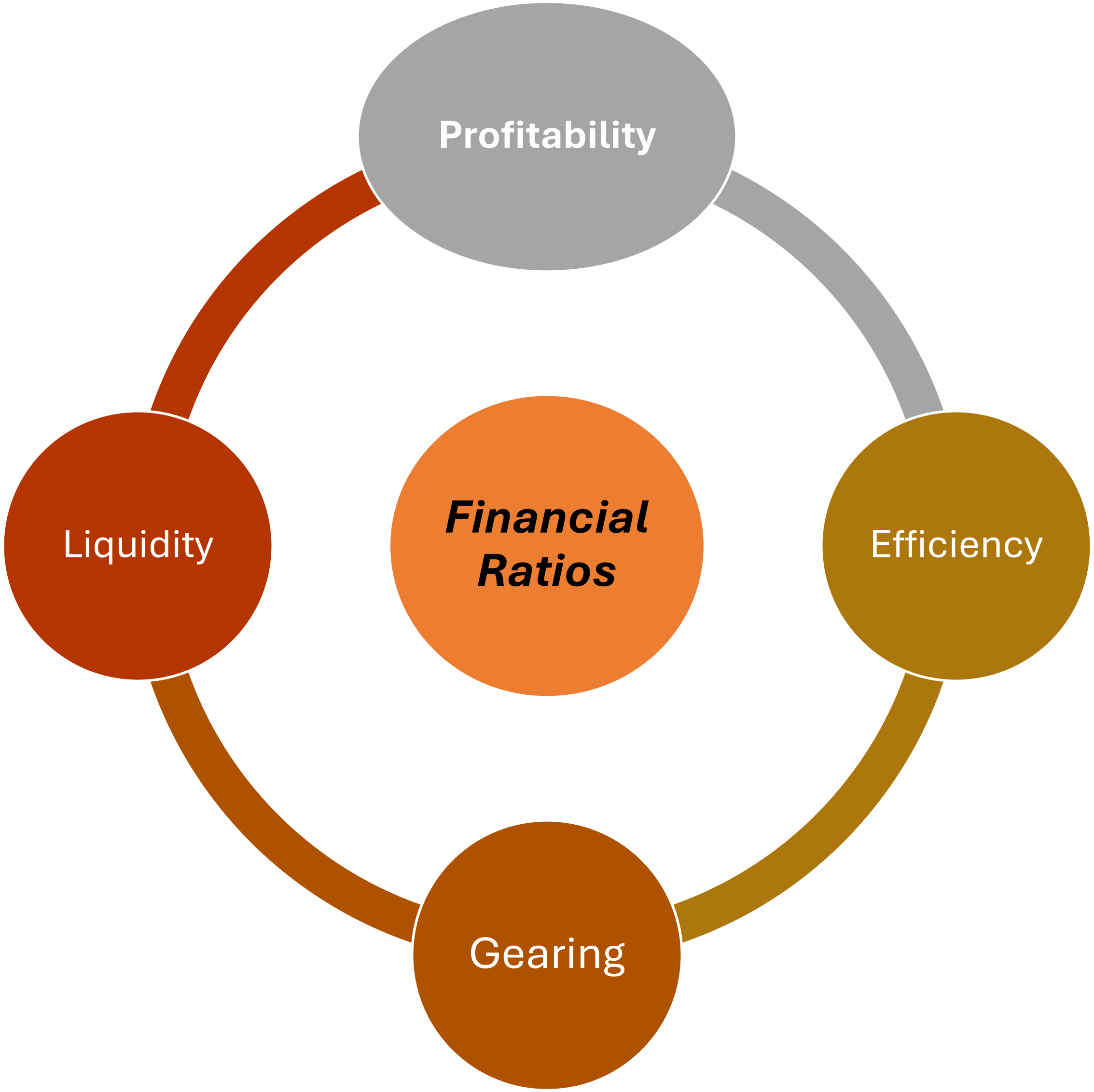
Compare Ratios With

- *Previous years*
- *Similar businesses*
- *Budgets or forecasts*
- *Industry stats or norms*
- *Other related ratios*

Key Questions When Using Ratios

1. What does the ratio tell us?
2. Has it changed—if so, why?
3. What is the expected benchmark?
4. What are its limitations?

Types of Financial Ratios



Profitability Ratios

"Is the business earning enough from what it does?"

→ Show how well a company makes money from its activities.

Efficiency Ratios

"Is the company getting good value from what it uses?"

→ Measure how effectively resources like stock or staff are used.

Liquidity Ratios

"Can it cover what it owes soon without running into trouble?"

→ Reveal how easily a company can pay short-term bills.

Gearing Ratios

"Is the business balanced between debt and its own money?"

→ Indicate how much of the company is funded by borrowing.

Example: *Sage Plc*

Consolidated income statement
For the year ended 30 September 2024

	Note	2024 £m	2023 £m
Revenue	2.1, 3.1	**2,332**	2,184
Cost of sales		**(168)**	(156)
Gross profit		**2,164**	2,028
Selling and administrative expenses		**(1,712)**	(1,713)
Operating profit	2.2, 3.2, 3.3, 3.6	**452**	315
Finance income	3.5	**19**	12
Finance costs	3.5	**(45)**	(45)
Profit before income tax		**426**	282
Income tax expense	4	**(103)**	(71)
Profit for the year		**323**	211
Profit attributable to:			
Owners of the parent		**323**	211
Earnings per share attributable to the owners of the parent (pence)			
Basic	5	**32.10p**	20.75p
Diluted	5	**31.55p**	20.43p

All operations in the year relate to continuing operations.

Example: *Sage Plc*

Consolidated balance sheet
As at 30 September 2024

	Note	2024 £m	2023 £m
Non-current assets			
Goodwill	6.1	**2,130**	2,245
Other intangible assets	6.2	**219**	274
Property, plant and equipment	7	**108**	104
Equity investments		**6**	4
Trade and other receivables	8.1	**137**	138
Deferred income tax assets	11	**81**	56
Derivative financial instruments	13.5	**29**	1
		2,710	2,822
Current assets			
Trade and other receivables	8.1	**404**	376
Current income tax asset		**16**	42
Cash and cash equivalents (excluding bank overdrafts)	12.3	**508**	696
		928	1,114
Total assets		**3,638**	3,936
Current liabilities			
Trade and other payables	8.2	**(405)**	(378)
Current income tax liabilities		**(26)**	(25)
Borrowings	12.4	**(15)**	(14)
Provisions	9	**(22)**	(23)
Deferred income	8.3	**(758)**	(745)
		(1,226)	(1,185)
Non-current liabilities			
Borrowings	12.4	**(1,231)**	(1,243)
Post-employment benefits	10	**(23)**	(19)
Deferred income tax liabilities	11	**(18)**	(18)
Provisions	9	**(25)**	(24)
Trade and other payables	8.2	**(3)**	(13)
Deferred income	8.3	**(6)**	(7)
Derivative financial instruments	13.5	**(13)**	(20)
		(1,319)	(1,344)
Total liabilities		**(2,545)**	(2,529)
Net assets		**1,093**	1,407
Equity attributable to owners of the parent			
Ordinary shares	14.1	**11**	12
Share premium		**548**	548
Other reserves	14.3	**88**	189
Retained earnings	14.4	**446**	658
Total equity		**1,093**	1,407

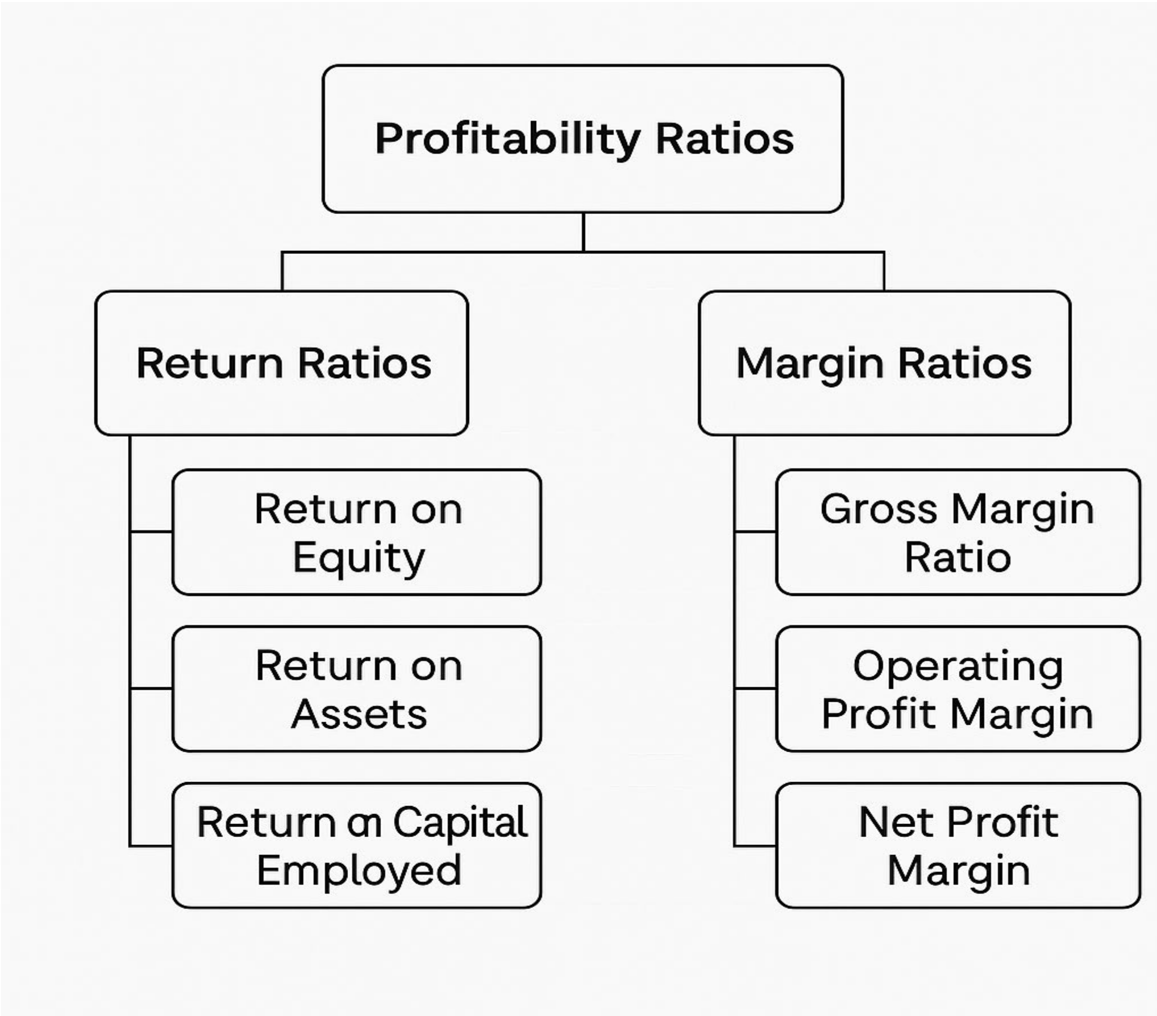
The consolidated financial statements on pages 175 to 252 were approved by the Board of Directors on 19 November 2024 and are signed on their behalf by:

J. H. Howell

Jonathan Howell
Chief Financial Officer

Strategic Report
Governance Report
Financial Statements
Additional Information

Profitability Ratios



- **Return Ratios**

  Show how well a business generates returns for shareholders, using balance sheet comparisons.

- **Margin Ratios**

  Indicate how much profit a company earns from its revenue, using income statement figures.

Profitability Return Ratios: Formulas

$$ROE = \frac{\text{Net Income}}{\text{Shareholders' Equity}}$$

$$ROA = \frac{\text{Net Income}}{\text{Total Assets}}$$

$$ROCE = \frac{\text{Operating Profit}}{\underbrace{\text{Total Assets} - \text{Current Liabilities}}_{\text{Capital Employed}}}$$

Profitability Margin Ratios: Formula

$$\text{Gross Margin Ratio} = \frac{\overbrace{\text{Total Revenue} - \text{COGS}}^{\text{Gross Profit}}}{\text{Total Revenue}}$$

$$\text{Operating Profit Margin} = \frac{\overbrace{\text{Total Revenue} - \text{COGS} - \text{Operating Expenses} - \text{Depreciation \& Amortization}}^{\text{Operating Profit or EBIT}}}{\text{Total Revenue}}$$

$$\text{Net Profit Margin} = \frac{\text{Net Income}}{\text{Total Revenue}}$$

Profitability Ratios: *Interpretation*

Ratio	What It Measures	Formula (Basic)	Interpretation Highlights
Return on Equity (ROE)	Profit earned per unit of shareholders’ equity	Net Income ÷ Shareholders’ Equity	Indicates how well owners' funds are being used; higher ROE suggests strong returns
Return on Assets (ROA)	Profit generated from total assets	Net Income ÷ Total Assets	Assesses efficiency of asset use; varies by industry type
Return on Capital Employed (ROCE)	Profit earned per unit of capital invested	Operating Profit ÷ Capital Employed	Highlights how well a business turns capital into profits; higher = more efficient
Gross Margin	Profit after covering production costs	(Revenue – COGS) ÷ Revenue	Shows basic profitability; high margin = better pricing or production efficiency
Operating Profit Margin	Profit from core operations before interest & tax	Operating Profit ÷ Revenue	Reflects how efficiently the company manages operating costs
Net Profit Margin	Bottom-line profitability after all expenses	Net Income ÷ Revenue	Measures the percentage of total revenue that becomes net profit

Profitability Ratios: *Example - Sage Plc*

Ratios		Formula	2024		2023		Interpretation
Profitability Return Ratios	**Return on Assets (ROA)**	Net Income / Total Assets	£323/£3,638	**8.88%**	£211/£3,936	**5.36%**	Return on Assets saw a significant increase in 2024, indicating improved efficiency in using assets to generate profit. For every pound of assets, the company generated 8.88 pence in profit in 2024, up from 5.36 pence in 2023.
	Return on Equity (ROE)	Net Income / Shareholders Equity	£323/£1,093	**29.55%**	£211/£1,407	**15.00%**	Return on Equity nearly doubled in 2024, showing a substantial improvement in profitability from shareholders' equity. This suggests that Sage Plc generated a higher profit for each pound of equity invested by its owners.
	Return on Capital Employed (ROCE)	Operating Profit / Capital Employed	£452/(£3,638-£1,226)	**18.74%**	5/(£3,936-£1,1	**11.45%**	The Return on Capital Employed increased significantly in 2024. This indicates that the company has become more effective at using its long-term financing to generate operating profits.

Profitability Ratios: *Example - Sage Plc*

Ratios		Formula	2024		2023		Interpretation
Profitability Margin Ratios	**Gross Profit Margin**	Gross Profit / Revenue	£2,164/£2,332	**92.80%**	£2,028/£2,184	**92.86%**	The gross profit margin remained exceptionally high and stable in both 2023 and 2024. This suggests that the direct cost of sales is very low relative to revenue, which is characteristic of a software and services business model.
	Operating Profit Margin	Operating Profit / Revenue	£452/£2,332	**19.38%**	£315/£2,184	**14.42%**	Sage Plc's operating profit margin increased significantly from 14.4% in 2023 to 19.4% in 2024. This improvement indicates greater operational efficiency and better management of selling and administrative expenses relative to the revenue generated.
	Net Profit Margin	Net Profit / Revenue	£323/£2,332	**13.85%**	£211/£2,184	**9.66%**	The net profit margin saw a substantial rise from 9.7% in 2023 to 13.9% in 2024. This growth reflects the improved operating performance and shows that a larger portion of revenue was converted into profit for the company's shareholders in 2024

Gearing/Leverage Ratios

What They Show

*How much a company relies on **borrowed funds** versus its own money (equity) to finance operations.*

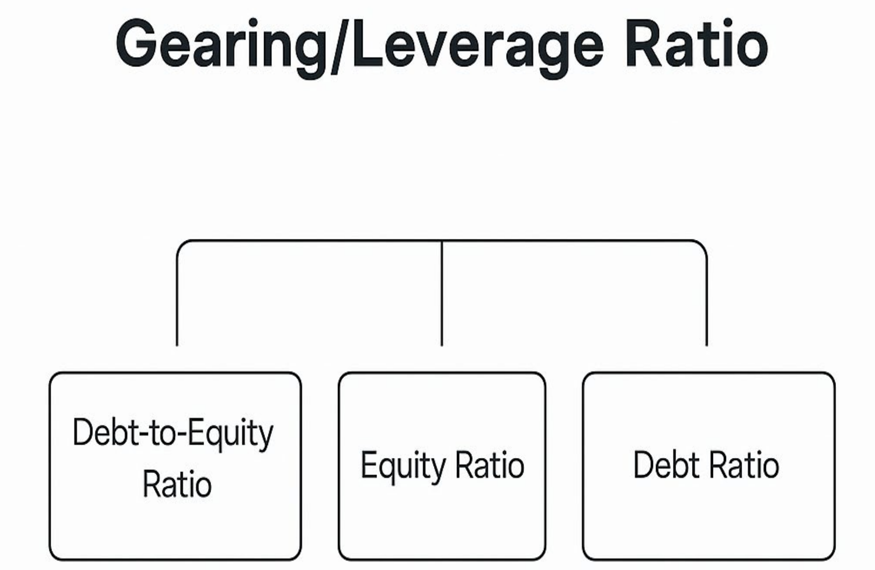
Why They Matter:

- Assess **financial risk** and long-term stability
- Reveal whether a business is **heavily debt-dependent** or conservatively financed
- Help investors and lenders judge **solvency and resilience**

Key Insight:

➡ High leverage = higher risk but potential for higher return

➡ Low leverage = lower risk, but may limit growth potential



Gearing Ratios: Formula

$$\text{Debt-to-Equity Ratio} = \frac{\text{Total Liabilities}}{\text{Shareholders' Equity}}$$

$$\text{Equity Ratio} = \frac{\text{Shareholders' Equity}}{\text{Total Assets}}$$

$$\text{Debt Ratio} = \frac{\text{Total Liabilities}}{\text{Total Assets}}$$

Gearing Ratios: *Interpretation*

Ratio	What It Measures	Formula (Basic)	Interpretation Highlights
Debt-to-Equity	Debt used for every unit of equity	Total Debt ÷ Shareholders' Equity	High ratio = more borrowing; suitable for stable cash flow firms but riskier overall
Equity Ratio	Portion of assets financed by shareholders' equity	Shareholders' Equity ÷ Total Assets	High ratio = strong financial independence; more owned than owed
Debt Ratio	Portion of assets financed through debt	Total Debt ÷ Total Assets	Ratio ≥ 1 = high leverage and risk; low ratio = more conservative capital structure

Gearing Ratios: *Example - Sage Plc*

Ratios		Formula	2024		2023		Interpretation
Gearing Ratios	**Debt-to-Equity**	Total Liabilities / Shareholders' Equity	£2,545/£1,093	**2.33**	£2,529/£1,407	**1.80**	company is using more debt to finance its assets, which can imply higher financial risk. Sage Plc's debt-to-equity ratio increased from approximately 1.80 in 2023 to 2.33 in 2024. This suggests that the company is relying more on debt financing relative to equity in 2024
	Equity	Shareholders' Equity / Total Assets	£1,093/£3,638	**0.30**	£1,407/£3,936	**0.36**	The equity ratio indicates the proportion of total assets financed by shareholders' equity. A higher equity ratio generally implies a stronger financial position as the company is less reliant on debt. Sage Plc's equity ratio decreased from approximately 0.36 in 2023 to 0.30 in 2024, indicating a lower proportion of assets financed by equity in 2024.
	Debt	Total Liabilities / Total Assets	£2,545/£3,638	**0.70**	£2,529/£3,936	**0.64**	The debt ratio shows the proportion of a company's assets that are financed by debt. A higher debt ratio suggests greater financial risk. Sage Plc's debt ratio increased from approximately 0.64 in 2023 to 0.70 in 2024. This indicates that a larger portion of the company's assets were financed by debt in 2024 compared to 2023, aligning with the trend observed in the debt-to-equity and equity ratios.

Liquidity Ratios

What They Show

Liquidity ratios reveal a company's ability to meet its short-term financial obligations using its most liquid assets. These ratios help determine how quickly a company can convert assets to cash without significantly affecting their value.

Why They Matter

1. **Creditworthiness** – Lenders and creditors use them to assess whether the business can repay loans on time.
2. **Operational Efficiency** – Strong liquidity suggests a healthy cash flow cycle and solid financial management.
3. **Risk Assessment** – Helps investors gauge how vulnerable a firm is to economic disruptions or market shifts.

Ratio	Insight Gained
Current Ratio	Measures if current assets cover current liabilities. A ratio >1 suggests liquidity.
Quick Ratio	Excludes inventories for a stricter test. Highlights if a firm can meet obligations fast.
Cash Ratio	Focuses solely on cash and cash equivalents—ultimate test of liquidity strength.

Liquidity Ratios: *Formular*

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

$$\text{Quick Ratio} = \frac{(\text{Current Assets} - \text{Inventory})}{\text{Current Liabilities}}$$

$$\text{Cash Ratio} = \frac{\text{Cash and Cash Equivalents}}{\text{Current Liabilities}}$$

Liquidity Ratios: *Interpretation*

Ratio	What It Measures	Interpretation Highlights
Current Ratio	Ability to meet short-term obligations using all current assets	✅ A ratio >1 suggests financial health. 🔄 Includes inventory, receivables, cash, etc. ⚠ Too high may indicate excess idle assets not being invested for growth.
Quick Ratio	Ability to meet short-term obligations using only most liquid assets (excl. inventory)	✅ A ratio >1 shows strong liquidity. 💧 Focuses on cash, marketable securities, receivables. ⚠ Very high levels may indicate inefficient use of cash reserves.
Cash Ratio	Ability to pay short-term liabilities using only cash and cash equivalents	📉 More conservative than other liquidity ratios. 👍 Ratio between 0.5–1 preferred. ⚠ Too high can signal under-utilized cash that could be used to fuel business growth.

Liquidity Ratios: *Example - Sage Plc*

	Ratios	Formula	2024		2023		Interpretation
Liquidity Ratios	**Current Ratio**	Current Assets / Current Liabilities	£928/£1,226	**0.76**	£1,114/£1,185	**0.94**	Sage Plc's current ratio decreased from approximately 0.94 in 2023 to 0.76 in 2024. A current ratio below 1 suggests that a company may not have enough liquid assets to cover its short-term obligations. While a ratio below 1 might raise concerns, for software companies with high deferred revenue (which is a current liability) and recurring revenue models, this can be less of a concern than for manufacturing or retail businesses. The decrease indicates a weakening in short-term liquidity from 2023 to 2024
	Quick Ratio	Current Assets minus Inventory / Current Liabilities	£928 - £0 / £1,226	**0.76**	£1,114 - £0 / £1,185	**0.94**	Sage Plc's quick ratio decreased from approximately 0.94 in 2023 to 0.78 in 2024. Similar to the current ratio, a quick ratio below 1 indicates that without relying on inventory (which is not present for Sage), the company's highly liquid assets are less than its current liabilities. The trend shows a decline in immediate liquidity
	Cash Ratio	Cash and Cash Equivalents / Current Liabilities	£508/£1,226	**0.41**	£696/£1,185	**0.59**	Sage Plc's cash ratio decreased from approximately 0.59 in 2023 to 0.41 in 2024. This indicates that the company's cash and cash equivalents alone could cover about 59% of its current liabilities in 2023, which dropped to 41% in 2024. This decline suggests a weaker ability to cover short-term obligations with readily available cash.

Efficiency Ratios

🔎 **What They Show**

- How well a company turns resources (assets, inventory, receivables) into sales or cash
- The speed and effectiveness of core operations

💡 **Why They Matter**

- Reveal operational performance and working capital health
- Help identify cash flow timing, inventory control, and customer payment habits
- Assist in benchmarking against competitors or industry standards

📌 **Key Insights**

- High turnover = faster collections or inventory movement (generally better)
- Long collection or inventory days = possible cash flow bottlenecks
- Must be interpreted in industry context and alongside profitability ratios

Efficiency Ratios: *Formula*

$$\text{Accounts Receivable Turnover Ratio} = \frac{\text{Net Credit Sales}}{\text{Average Accounts Receivable}}$$

Interpretation

To calculate this ratio, the following formulas are also necessary:

Net Credit Sales = Sales on Credit – Sales Returns – Sales Allowances

$$\text{Average Accounts Receivable} = (\text{Accounts Receivable}_{ending} + \text{Accounts Receivable}_{beginning})/2$$

$$\text{Asset Turnover Ratio} = \frac{\text{Net Sales}}{\text{Average Total Assets}}$$

Interpretation

To calculate this ratio, average total assets is calculated as:

$$\text{Average Total Assets} = (\text{Total Assets}_{ending} + \text{Total Assets}_{beginning})/2$$

$$\text{Inventory Turnover Ratio} = \frac{\text{Cost of Goods Sold}}{\text{Average Inventory}}$$

Interpretation

To calculate this ratio, average inventory is calculated as:

$$\text{Average Inventory} = (\text{Inventory}_{ending} + \text{Inventory}_{beginning})/2$$

Efficiency Ratios: *Interpretation*

Efficiency Ratio	What It Measures	Formula (Basic)	Interpretation Highlights
Accounts Receivable Turnover	How often receivables are collected during a period	Credit Sales ÷ Avg. Accounts Receivable	High = faster collection; low = possible cash flow issues
Asset Turnover Ratio	Efficiency of asset use to generate revenue	Revenue ÷ Total Assets	High = better use of assets; varies by industry
Inventory Turnover Ratio	How often inventory is sold and replaced in a period	Cost of Goods Sold ÷ Avg. Inventory	High = efficient stock movement; low = overstocking or slow sales

Efficiency Ratios: *Example - Sage Plc*

Here are the actual figures for 2024 from Sage Plc financial statements:

- **Revenue (2024):** £2,332m
- **Cost of Sales (2024):** £168m
- **Trade and other receivables - Current assets (2024):** £404m
- **Trade and other receivables - Non-current assets (2024):** £137m
- **Total Assets (2024):** £3,638m

To calculate the *"Average Accounts Receivable"* and *"Average Total Assets"* for 2024, we also need the 2023 figures:

- **Trade and other receivables - Current assets (2023):** £376m
- **Trade and other receivables - Non-current assets (2023):** £138m
- **Total Assets (2023):** £3,936m

Now, let's assume a figure for inventory *(There is no inventory in Sage Plc financial statement*). Since Sage Plc is primarily a software company, inventory would likely be low.

- **Assumed Ending Inventory (2024):** £10m
- **Assumed Beginning Inventory (2023 - for average 2024):** £8m

Efficiency Ratios: *Example - Sage Plc*

Efficiency Ratio Calculations for Sage Plc (2024)

First, we calculate the average values needed for the ratios:

1. Total Trade and Other Receivables:

- **2023:** £376m (current) + £138m (non-current) = £514m
- **2024:** £404m (current) + £137m (non-current) = £541m

2. Average Accounts Receivable (2024):

$$\frac{\text{Ending AR 2023} + \text{Ending AR 2024}}{2} = \frac{514 + 541}{2} = \frac{1,055}{2} = 527.5\text{m}$$

3. Average Inventory (2024):

$$\frac{\text{Assumed Beginning Inventory 2024} + \text{Assumed Ending Inventory 2024}}{2} = \frac{8 + 10}{2} = \frac{18}{2} = 9\text{m}$$

4. Average Total Assets (2024):

$$\frac{\text{Ending Total Assets 2023} + \text{Ending Total Assets 2024}}{2} = \frac{3,936 + 3,638}{2} = \frac{7,574}{2} = 3,787\text{m}$$

Efficiency Ratios: *Example - Sage Plc*

1. Accounts Receivable Turnover Ratio

This ratio indicates how efficiently a company collects its outstanding receivables.

$$\text{Accounts Receivable Turnover} = \frac{\text{Revenue}}{\text{Average Accounts Receivable}}$$

$$\frac{2,332\text{m}}{527.5\text{m}} \approx 4.42 \text{ times}$$

Interpretation: Sage Plc's Accounts Receivable Turnover for 2024 is approximately 4.42 times. This suggests that the company collected its average receivables about 4.42 times during the year, indicating a reasonably efficient collection process.

2. Asset Turnover Ratio

This ratio measures how efficiently a company is using its assets to generate sales.

$$\text{Asset Turnover} = \frac{\text{Revenue}}{\text{Average Total Assets}}$$

$$\frac{2,332\text{m}}{3,787\text{m}} \approx 0.62 \text{ times}$$

Interpretation: Sage Plc's Asset Turnover for 2024 is approximately 0.62 times. This means that for every £1 of assets, Sage Plc generated about £0.62 in revenue. This ratio can vary significantly by industry, but it provides insight into how effectively the company is using its asset base to generate sales.

Efficiency Ratios: *Example - Sage Plc*

3. Inventory Turnover Ratio

This ratio indicates how many times a company has sold and replaced its inventory during a period.

$$\text{Inventory Turnover} = \frac{\text{Cost of Sales}}{\text{Average Inventory}}$$

$$\frac{168\text{m}}{9\text{m}} \approx 18.67 \text{ times}$$

Interpretation: Sage Plc's Inventory Turnover for 2024 is approximately 18.67 times (based on assumed inventory figures). A high inventory turnover ratio, especially for a software company where physical inventory is minimal, generally indicates efficient inventory management and strong sales relative to the inventory held.

Limitations of Ratio Analysis

1. **Based on Historical Data**

   Reflects past performance — may not predict future trends.

2. **Inflation Distortion**

   Financials from different periods may not be comparable unless inflation adjusted.

3. **Changes in Accounting Policies**

   Policy shifts can affect consistency in metrics across periods.

4. **Operational Restructuring**

   Major business changes can make before-and-after comparisons misleading.

5. **Seasonality**

   Failing to account for seasonal patterns may distort interpretation.

6. **Potential for Manipulation**

   Financial data can be misrepresented; users must apply due diligence.

Thank you, very much!

Any questions?

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References

Corporate Finance Institute. (n.d.) *CFI financial ratios eBook*. Available at: https://corporatefinanceinstitute.com/assets/CFI-Financial-Ratios-Cheat-Sheet-eBook.pdf

Corporate Finance Institute (n.d.) *Limitations of ratio analysis*. Available at: https://corporatefinanceinstitute.com/resources/accounting/limitations-ratio-analysis/

Sage Group plc. (2024) *Annual report and accounts 2024*. Available at: https://www.sage.com/en-gb/-/media/files/investors/documents/pdf/annual%20report/sage-annual-report-2024.pdf